ALEX ANTONY

Software Engineer | Python Developer | AI/ML Engineer | Full-Stack Developer
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PROFESSIONAL SUMMARY

MCA graduate and aspiring Software Engineer with strong Python programming skills and hands-on experience building production-ready AI and full-stack applications. Developed a bilingual Retrieval-Augmented Generation (RAG) based mobile learning app with spaced repetition (FastAPI, React Native, ChromaDB), an AI-powered Voice Interview Coach (LangChain, FastAPI, React), a YOLOv8-based real-time traffic management system, and a Django-based service management platform. Skilled in Software Development Life Cycle (SDLC), Object-Oriented Programming (OOP), RESTful API design, Data Structures and Algorithms (DSA), and full-stack web development. Strong mathematical foundation (B.Sc. Mathematics) supporting algorithmic thinking and problem solving. Actively solving Data Structures and Algorithms (DSA) problems on LeetCode. Seeking a Software Engineer role to contribute to scalable, high-quality software products.

TECHNICAL SKILLS

Programming Languages: Python (primary), JavaScript, TypeScript, Java, C++

Software Development: Object-Oriented Programming (OOP), REST API Design, Software Development Life Cycle (SDLC), Agile, Unit Testing, Integration Testing, System Testing, Version Control (Git)

Web & Backend Frameworks: FastAPI, Django, Flask, React.js, React Native, HTML, CSS, Bootstrap

AI / ML & Generative AI: LangChain, Groq API, Retrieval-Augmented Generation (RAG), Vector Databases (ChromaDB), Prompt Engineering, YOLOv8 (PyTorch), scikit-learn, NumPy, Pandas, Matplotlib, OpenCV

Databases: MySQL, SQLite, ChromaDB, Relational Database Design

Tools & Platforms: Git, GitHub, Linux, Docker (basic), VS Code, Postman, Expo

Core Concepts: Data Structures & Algorithms (DSA), Problem Solving, System Design (basics), Statistics, Probability, Linear Algebra

EDUCATION

Master of Computer Applications (MCA) *2024 - 2026*
College of Engineering, Chengannur - APJ Abdul Kalam Technological University | CGPA: 8.26

Bachelor of Science in Mathematics *2020 - 2024*
SB College, Changanacherry - Mahatma Gandhi University | CGPA: 7.0

PROJECTS

Second Brain - AI-Powered Learning App with RAG & Spaced Repetition | GitHub *Aug 2026*

- Built a bilingual (English/Malayalam) mobile learning app using FastAPI, React Native (Expo), Groq LLM (GPT-OSS 120B), and ChromaDB for Retrieval-Augmented Generation (RAG) to deliver context-aware, personalized topic explanations with real-world analogies.
- Designed a custom spaced-repetition scheduling algorithm (SQLite-backed) implementing an SM-2-inspired interval system to optimize long-term retention of learned concepts.
- Implemented an active-recall self-test feature generating novel interview-style questions from prior learning history, distinct from initial study content, with user-response capture and model-answer comparison.
- Engineered a dual-LLM-call prompt architecture to solve prompt-length and topic-drift issues in low-resource-language (Malayalam) generation, improving output completeness and topic fidelity.
- Organized content into user-defined folders with cross-topic semantic search via vector embeddings; diagnosed and migrated off a deprecated LLM model in a live integration.

Tech Stack: *Python, FastAPI, React Native, Expo, Groq API, ChromaDB, SQLite, Prompt Engineering, RESTful API Design*

AI Interview Coach | GitHub *Jun 2026*

- Built a full-stack AI-powered voice interview simulator using Python, LangChain, Groq LLM (LLaMA 3.3 70B), FastAPI, and React.js, simulating a real interview experience with voice-to-voice interaction.
- Implemented continuous speech recognition (Web Speech API) with manual End Answer control, preventing auto-cutoff during natural pauses and replicating realistic interview conditions.
- Designed RESTful API endpoints using FastAPI for dynamic question generation across Technical, HR, and Mixed interview modes with Easy, Medium, and Hard difficulty levels.
- Built resume-based personalized question generation; the AI analyzes an uploaded PDF resume and generates questions specific to the candidate's projects, skills, and experience.
- Developed automated answer scoring (0-10) with structured feedback covering strengths, improvement areas, and better answer suggestions, along with session progress tracking via visual analytics.

Tech Stack: *Python, LangChain, Groq API, FastAPI, React.js, Web Speech API, REST API*

Smart Traffic Management System (MCA Final Year Project) | GitHub *Mar 2026*

- Designed and implemented a real-time vehicle detection and classification system using the YOLOv8 deep learning model, processing live video streams with 90%+ detection accuracy.
- Built a complete data preprocessing pipeline using OpenCV, including frame extraction, resizing, normalization, and enhancement for varying lighting and traffic conditions.
- Developed a RESTful backend using Flask with model inference and video streaming endpoints; built a React.js dashboard for live monitoring and visualization.
- Implemented an adaptive traffic signal control algorithm (RED/YELLOW/GREEN phases) driven by real-time vehicle density estimation, reducing manual intervention.
- Conducted comprehensive testing (unit, integration, system) and evaluated the model using Precision, Recall, F1-score, and mAP metrics.

Tech Stack: *Python, YOLOv8 (PyTorch), OpenCV, Flask, React.js, NumPy, Pandas*

GameHub - Mobile Game Discovery App | GitHub *Apr 2026*

- Built a cross-platform mobile application using React Native (Expo) with TypeScript, targeting both Android and iOS from a single codebase.
- Implemented file-based routing using Expo Router and structured the project layout following mobile development best practices.

Tech Stack: *React Native, Expo, TypeScript, JavaScript, Expo Router*

Autocare - Vehicle Service Management System | GitHub *Oct 2025*

- Developed a multi-role web application (Customer, Mechanic, Manager, Admin) with complete authentication, authorization, and role-based access control (RBAC) using Django.
- Designed a normalized relational database schema in MySQL; implemented MVC architecture with a modular code structure following software engineering best practices.

Tech Stack: *Django, MySQL, Bootstrap, Python*

CERTIFICATIONS

- SWAYAM (NPTEL) - Cloud Computing

ACTIVITIES & INTERESTS

- Actively solving Data Structures and Algorithms (DSA) problems on LeetCode covering arrays, strings, trees, graphs, and dynamic programming, focused on technical interview preparation.
- Building personal AI projects to stay current with LLM and Generative AI (GenAI) developments.